Optical Service System Proposal

Improve eye care service and user convenience



HNDIT4032

Name of the Supervisor:

Mr. Thilina Bandara

Department of Information Technology

Advanced Technological Institute

Rathnapura

W.P.M.Malindi Pabasara

RAT/IT/2324/F/0018

Contents

Page

1. Introduction / Background…… 3
2. Problem Statement…… 4
3. Objectives……. 5
    - 3.1 General Objectives……. 5
    - 3.2 Specific Objectives……. 5
4. Proposed Solution……. 6
5. Methodology / Development Approach…… 7
6. System Requirements…… 8
    - 6.1 Functional Requirements…… 8
    - 6.2 Non-Functional Requirements……. 8
7. Tools and Technologies……. 9
8. Project Timeline / Schedule……... 10
9. Expected Outcomes (Deliverables, Benefits)…… 11
10. Risks and limitations……. 12
11. Conclusion……. 13

1. Introduction / Background

The optical service industry plays a vital role in maintaining eye health and providing vision correction solutions such as prescription glasses, contact lenses, and eye testing services. Traditionally, many optical service providers rely on manual systems to manage customer records, prescriptions, and product sales. These outdated methods often lead to inefficiencies, errors, and delays in service delivery, affecting both the business and customer satisfaction.

With the rapid advancement of digital technology, there is a growing demand for modern, web-based solutions that can improve the efficiency and accessibility of optical services. Customers today prefer online platforms where they can easily book appointments, access their prescription history, and browse available products from the comfort of their homes. Optical businesses also need better tools to manage operations, track inventory, and communicate effectively with their customers.

This project proposes the development of a web-based Optical Service System designed to address these challenges. The system will provide an integrated platform for managing appointments, customer records, and product sales while offering a user-friendly experience. By digitizing optical services, the system aims to enhance operational efficiency, reduce errors, and improve overall customer satisfaction.

2. Problem Statement

Many optical service businesses still rely on manual or semi-digital processes to manage customer appointments, eye check records, frame selection, and order tracking. This often leads to issues such as long waiting times, miscommunication, lost customer records, and difficulty in managing inventory efficiently. Customers also face challenges in booking appointments, checking available products, and receiving timely updates about their orders. As a result, both service providers and customers experience reduced efficiency and satisfaction in the overall service process.

- ✓ Traditional optical service systems face several problems:
- ✓ Manual record keeping leads to data loss and errors
- ✓ Difficulty in managing prescriptions and customer history
- ✓ No online booking system for eye testing
- ✓ Limited access for customers to browse and order products
- ✓ Poor communication between customers and optical stores

3. Objectives

3.1 General Objective

To develop a web-based optical service system that digitizes and improves the management of optical services.

3.2 Specific Objectives

- To create an online platform for booking eye test appointments
- To manage customer records and prescriptions digitally
- To provide an online store for glasses and lenses
- To enable secure payment and order tracking
- To improve communication between customers and service providers

4. Proposed Solution

The proposed system is a web-based Optical Service Management System that allows:

- ✓ Customers to register and log in
- ✓ Book eye test appointments online
- ✓ View prescriptions and history
- ✓ Browse and purchase optical products
- ✓ Admin to manage products, users, and appointments
- ✓ This system will reduce manual work and improve service efficiency.

5. Methodology / Development Approach

The system will be developed using the Agile methodology, allowing continuous improvement and feedback.

Steps include:

- Requirement gathering
- System design (UI/UX, database)
- Development (frontend & backend)
- Testing (unit and system testing)
- Deployment and maintenance

6. System Requirements

6.1 Functional Requirements

- User registration and login
- Appointment booking system
- Prescription management
- Product catalog (glasses, lenses)
- Shopping cart and checkout
- Admin dashboard
- Order tracking system

6.2 Non-Functional Requirements

- User-friendly interface
- Fast performance
- Secure data handling
- Responsive design (mobile + desktop)
- High system availability

7. Tools and Technologies

- Frontend: HTML, CSS, JavaScript (React or Bootstrap)
- Backend: Node.js / PHP / Java (Spring Boot)
- Database: MySQL / MongoDB
- Version Control: Git
- IDE: VS Code / IntelliJ

8. Project Timeline / Schedule

Phase	Duration
Requirement Analysis	1 week
Design	1 week
Development	3 week
Testing	1 week
Deployment	1 week

9. Expected Outcomes (Deliverables, Benefits)

Deliverables

- Fully functional web system
- Database system
- Documentation
- User manual

Benefits

- Easy access to optical services
- Reduced manual errors
- Improved customer satisfaction
- Efficient management system

10. Risks and Limitations

The development and implementation of the Optical Service Web System may face several risks and limitations that could affect its performance and usability. Technical risks such as system bugs, server downtime, and data security vulnerabilities may arise if the system is not properly tested and maintained. There may also be limitations related to internet dependency, as users need stable connectivity to access the system efficiently. Additionally, some staff members may require training to adapt to the new digital system, which could temporarily slow down operations. Budget constraints and time limitations during development can also impact the quality and full functionality of the system. Therefore, proper planning, regular maintenance, and user training are essential to minimize these risks and ensure the system operates successfully.

- ✓ Internet dependency
- ✓ Data security risks
- ✓ Limited user technical knowledge
- ✓ Time constraints during development

11. Conclusion

The Optical Service Web System provides an effective solution to overcome the challenges faced by traditional optical service management. By automating key processes such as appointment booking, customer record management, product handling, and order tracking, the system significantly improves operational efficiency and reduces the chances of human error. It also enhances communication between customers and service providers, ensuring a smoother and more organized service experience.

Furthermore, the implementation of this system will lead to higher customer satisfaction by offering convenience, accessibility, and faster service. It enables customers to interact with the optical service center anytime and from anywhere, while helping businesses manage their operations more professionally. Overall, the Optical Service Web System contributes to modernizing optical services and supports business growth through improved service quality and better resource management.